

David Kreuter

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Infrastructure & Networking Engineer | VPN, IP Management, Distributed Systems, and Security Automation

Infrastructure and networking engineer with 10+ years of experience building, securing, and scaling distributed systems in enterprise environments. Deep expertise in IP addressing, DNS architecture, VPN connectivity, firewall policy automation, and high-availability networking across geo-redundant data centers. Proven ability to diagnose complex networking issues at the hardware and protocol levels, architect secure hybrid-cloud solutions, and develop software that automates network operations at scale. Experienced with Python, SQL, API-driven orchestration, and the creation of observable, resilient services for high-stakes financial infrastructure.

WORK EXPERIENCE

M&T Bank

03/2019 - Present

Automation Engineer (formerly Systems Engineer)

- **Network Security Automation:** Automated Check Point firewall rule generation and DMZ provisioning; integrated security policy updates into the server build cycle to ensure day-zero network isolation and a hardened, security-first posture across all deployed infrastructure.
- **Operational Efficiency:** Developed an intelligent IP provisioning engine via InfoBlox API integrations--managing IP addressing, subnet allocation, and DNS registration at scale--eliminating 1,000+ annual engineering hours and resolving a multiyear infrastructure bottleneck.
- **Infrastructure Resilience:** Engineered and maintained a resilient distributed infrastructure spanning three identical, geo-redundant data centers, ensuring maximum uptime, disaster-recovery readiness, and consistent networking across sites.
- **Architectural Scale:** Engineered a global server provisioning framework that deployed and managed 20,000+ virtual machines; integrated Rubrik and VMware APIs to automate backups and compute placement, built for observability, and ensured 100% RTO/RPO compliance at build time.
- **Platform Engineering & Middleware:** Designed and developed a custom Python orchestration application as middleware between ServiceNow and Ansible Automation Platform; implemented advanced retry logic, auto-remediation of deployment failures, and a pub/sub queue model for high-volume scaling.
- **Cloud Abstraction & Redesign:** Refactored legacy on-premises automation workflows to be fully cloud-agnostic, enabling seamless deployments to Azure; designed the logic to be transparent to end users, maintaining a consistent interface while abstracting the underlying infrastructure provider.
- **Hybrid Cloud & GitOps:** Architected GitOps-driven CI/CD pipelines using Terraform to automate the deployment of virtual machines to Azure; implemented a pull-request-based workflow that standardized infrastructure changes through automated plan/apply cycles.
- **Database as a Service (DBaaS):** Collaborated with Database Engineering to architect a high-performance on-premises SQL cluster, enabling a unified DBaaS offering that consolidated fragmented legacy databases and streamlined application delivery.
- **End-to-End Hardening & Provisioning:** Developed a zero-touch build workflow that delivers fully patched and hardened Linux (Ansible) and Windows (GPO) servers; automated agent installation, drive formatting, and optional hardened SQL deployments at scale.
- **Risk Mitigation:** Authored Active Directory lifecycle scripts with built-in 'read-only' dry-run modes; implemented least-privilege permissions and robust rollback protocols for stale object cleanup.
- **Cost Optimization:** Orchestrated the migration of 850 branch file servers within a strict three-month window using custom PowerShell automation, avoiding \$400,000 in extended licensing fees.
- **Technical Leadership:** Serve as a primary mentor for the Technology Development Program (TDP), setting enterprise-wide standards for naming conventions, Git branching strategies, and unit testing for infrastructure code.

OTG Management

07/2013 - 02/2019

Lead Network Engineer

- **Infrastructure Resilience:** Architected a highly redundant DNS solution that significantly reduced network outages and optimized global traffic performance across distributed environments.
- **Cloud Connectivity:** Established secure, encrypted VPN tunnels to bridge on-premises networks with cloud resources, ensuring seamless hybrid-cloud operations and secure interconnectivity.
- **Security Architecture:** Engineered a secure reverse proxy solution to protect sensitive internal infrastructure, enabling customer online ordering while mitigating external threat vectors.
- **Performance Engineering:** Diagnosed and resolved a mission-critical server hang by identifying a NUMA architecture mismatch; realigned vCPU/memory topology to eliminate cross-node latency and restore system stability.

Network Engineer

- **Network Administration:** Managed complex firewall rules, routing policies, and switch ACLs for a distributed network spanning 12 airports and 3 satellite offices, diagnosing and resolving networking issues across diverse environments.
- **Cybersecurity Governance:** Directed enterprise-wide vulnerability scanning and remediation efforts for internal and external systems, maintaining a hardened security posture.

- **Proactive Monitoring:** Developed service-aware alerting and observability frameworks for business-critical systems, reducing mean time to resolution (MTTR) through proactive incident detection.
- **Enterprise Mobility:** Onboarded and managed 15,000 iPads via Meraki MDM, implementing automated security protocols including geo-fencing, remote-wipe triggers, and custom application deployment.

Systems Administrator

- **Disaster Recovery:** Spearheaded the deployment of Datto backup and disaster recovery appliances, ensuring 100% business continuity for critical operations.
- **Operational Visibility:** Modernized infrastructure monitoring by designing custom Grafana dashboards to visualize real-time performance metrics and system health.

SKILLS

Networking: VPN, DNS, BGP, IP Addressing & Management, High Availability, F5 Load Balancing, VRRP, Intrusion Detection, Vulnerability Scanning, Meraki MDM

Languages: Python, SQL, Bash, PowerShell, Javascript, YAML, PHP

Infrastructure: Check Point Firewalls, VMware vSphere, Azure Cloud Services, Infrastructure as Code (IaC), OS Hardening (CIS/STIG), Active Directory, DBaaS, Nutanix AHV, Rubrik, NUMA Tuning

Automation/Orchestration: REST API, Terraform, CI/CD Pipelines, Grafana, Ansible Automation Platform, Middleware Development, GitOps, Gitlab, ServiceNow, GraphQL, Unit Testing

EDUCATION

Bachelor Of Science in Computer Information Systems

12/2012

SUNY Fredonia

CERTIFICATIONS

Certified Meraki Network Operator

Meraki

VOLUNTEERING & LEADERSHIP

Habitat For Humanity

Volunteer

07/2025 - Present

Tifft Nature Preserve

Volunteer

01/2025 - Present

716 Paws

Thermal drone pilot for search and rescue missions.

01/2022 - Present